Loop theory describes this atomic and granular quantum structure of space in a precise mathematical form. It is obtained by applying the general equations of quantum mechanics written by Dirac to Einstein's gravitational field.

In particular, loop theory specifies that volume (for example, the volume of a given cube) cannot be arbitrarily small. A minimum volume exists. No space smaller than this minimum volume exists. There is a minimum quantum of volume: an elementary atom of space.

Atoms of space

Remember Achilles chasing after the tortoise? Zeno observed that there is something difficult to accept in the idea that Achilles has to cover an infinite number of distances before reaching the slow-moving creature. Mathematics had found a possible answer to this difficulty, showing how an infinite number of progressively smaller intervals could nevertheless amount to a finite total interval.

But is this what *truly* happens in nature? Are there intervals between Achilles and the tortoise that can be arbitrarily short? Does it really make sense to talk of a billionth of a billionth of a billionth of a millimetre, and then to think of dividing it again further innumerable times?

The calculation of the quantum spectra of geometric quantities indicates that the answer is negative: arbitrarily small chunks of space do not exist. There is a lower limit to the divisibility of space. It is at a very small scale indeed, but it is there. This is what Matvei Bronštejn had intuited in the 1930s. The calculation of the spectra of volume and area confirms Bronštejn's idea and frames it in a mathematically precise manner.

Achilles does not need an infinite number of steps to reach the tortoise because, in a space made of grains of finite size, infinitely small steps do not exist. The hero will come ever closer to the creature until, in the end, he reaches it in a single quantum leap.

But, on reflection, was this not precisely the solution proposed by Leucippus and by Democritus? They spoke of the granular structure of matter, and we are rather unsure as to what, precisely, they said about space. Unfortunately, we do not have their texts and must make do with the sparse fragments in the citations of others. It is like trying to reconstruct Shakespeare's plays from a list of Shakespeare quotes.[fn36] Democritus's argument on the incongruity of the continuum as a collection of points, reported by Aristotle, may be applied to space. I imagine that if we could ask Democritus if it makes sense to split a space interval ad infinitum, his reply could only be to repeat that divisibility must have a limit. For the philosopher of Abdera, matter is made of atoms that cannot be divided. Having once